DB4 Exam 2025

Note that these are not the correct answers but just my own ;)

Content **tion 1** (1 point) ✓ *Saved*

The PID controller used to regulate the temperature of the mussels tank is characterized by 3 terms: proportional, integral, and derivative. What is the purpose of the integral term?

- (●) Produce an action that is proportional to the integral of the error. Thus, reducing the steady state error.
- ( ) Integrate the overall PID output so that the PID response is smoother. In other words, it reduces fast variations of the PID output.
- ( ) Produce an action that is proportional to the rate of change of the error. Thus, dumping overshoot and oscillation.

Question 2 (1 point) ✓ *Saved*

The MQTT protocol is a lightweight messaging network protocol that:

- ( ) Is used to allow the ESP32 board to connect to the WiFi network without a password, still maintaining a certain level of security since the logged data (e.g., temperature, OD values, etc.) are stored remotely.
- (●) Is used to transfer data between the ESP32 board and the Adafruit IO server and it is based on a publish/subscribe approach.
- ( ) Is used by the ESP32 board to automatically download MicroPython firmware updates which include all the most recent network-related security patches.

Main Content n 3 (1 point) Saved

What is the purpose of the relay board used to drive the cooling module?

- (●) It allows to switch high-current/high-voltage loads with a digital signal produced by the ESP32 board. Such currents and voltages cannot directly be produced by the ESP32 board.
- ( ) It allows to safely switch high-current/high-voltage loads with a digital signal produced by the ESP32 board. Such high-currents and high-voltages can be also produced by the ESP32 board directly, but it is more reliable to use a relay.
- ( ) It allows to linearly control the power of high-current/high-voltage loads using an analog signal produced by the ESP32 board. More specifically, the power of the high-current/high-voltage load is directly proportional to the analog voltage provided by the ESP32 board.

Question 4 (1 point) Saved

What is the maximum flow rate of the large 12V Diaphragm Pump?

- ( ) 240 L/min
- ( ) 80-120 L/min
- ( ) 0.4-1.5 L/min
- (●) 2-3 L/min

Question 5 (1 point) ✓ *Saved*

The MicroPython programming language used to program the ESP32 board is an implementation of the Python language that:

- ○ Includes the entire set of the standard Python libraries and it is used for applications running on laptops or servers.
- ○ Includes Python libraries supporting very small font sizes and it is especially useful when using small displays.
- ● Includes a subset of the standard Python libraries and it is optimized to run on microcontrollers.

Content

Question 6 (1 point) ✓ *Saved*

What is the pumping capacity of the 3-6V small submersible Pump used for dosing in the project?

- ○ 240 L/h
- ● 80-120 L/h
- ○ 2-3 L/h
- ○ 1.25 L/h

Question 7 (1 point) ✓ *Saved*

How would you validate a pump?

- ○ Measure the pumped volume several time for one flow rate
- ● Measure the pumped volume for a given time period several times over a range of flow rates and measure pumped volumes with a measuring cylinder
- ○ Measure the pumped volume for a given time period over a range of flow rates and measure pumped volumes with a measuring cylinder

Question 8 (1 point) ✓ *Saved*

How are biocompatibility studies done?

- ◯ The cells are cultured together with the material and material degradation is studied
- ◉ Cells growth rate is compared between a test tube without and with the material to be tested
- ◯ Cells are cultured together with a material and time to proliferate is measured

Question 9 (1 point) ✓ *Saved*

How can heat losses be minimized in a system?

- ◯ Increase the flow rate of the cold water
- ◯ Cool the circulating water more
- ◉ Decrease the tube lengths and isolate the tubes

Question 10 (1 point) ✓ *Saved*

Which of the following statements can be considered as a milestone?

- ◯ "A method to build a device"
- ◯ "A protocol for growing algea"
- ◉ "We can grow algea in a bucket"

Main Content n **11** (1 point) ✓ *Saved*

What happens with you plankton feed algae growth rate, if the temperature drop from 20 to 10 degrees C?

- (●) Growth rate is halved
- ( ) They die
- ( ) Nothing

Question 12 (1 point) ✓ *Saved*

What method gives the most accurate estimate of you feed algae concentration if you have 20000 cells*ml^{-1} in your culture?

- ( ) Flourometer
- (●) Particle counter
- ( ) OD

Question 13 (1 point) ✓ *Saved*

At what light intensity (μE*m^{-2}*s^{-1}) do the feed algae grow best? (unit 'E' is Einstein)

- ( ) 1000
- ( ) 10
- (●) 100

Question 14 (1 point) ✓ *Saved*

At what cell concentration do you get the highest mussel clearance rate?

- ◯ 10000 cells*ml^{-1}
- ◉ 1000 cells*ml^{-1}
- ◯ 100000 cells*ml^{-1}

Question 15 (1 point) ✓ *Saved*

At what cell concentration do you get the highest mussel ingestion?

- ◯ 200000 cells*ml^{-1}
- ◉ 20000 cells*ml^{-1}
- ◯ 2000 cells*ml^{-1}